ARM-450 — Parts, Drawings and Dimensions

- Source: ~/Desktop/Robotic_arm_design/ (Fusion 360 export)
- Date: 2026-08-13
- Units: all dimensions in millimetres
- Method: every dimension below was measured directly from the STL geometry, not read off a drawing. Envelope = axis-aligned bounding box of the part as exported.

1. How to read this document

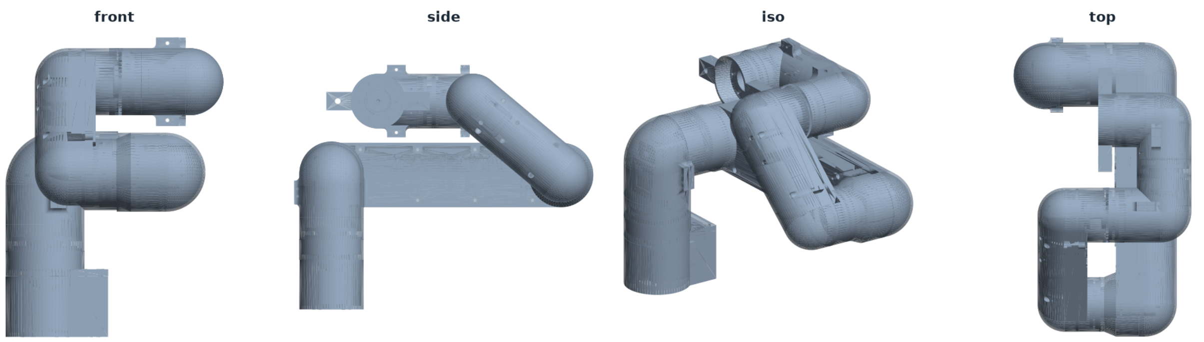
Each part carries a part number, its measured envelope, an estimated printed mass, and a status. Status meanings:

- keep — Carry forward unchanged.
- revise — Minor revision — fillets and bearing seats.
- LENGTHEN — Lengthen to 145 mm joint-to-joint.
- MISMATCH — Envelope does not pair with its base shell — check before reuse.
- REJECT — Unequal link length; creates a dead zone and exceeds 450 mm.
- REDESIGN — Redesign as a full-perimeter collar; current form is understrength.
- FAILS — Fails at servo stall torque. Do not reprint as-is.
- COTS — Commercial off-the-shelf, not printed.
- REPLACE — Replace with a spaced, preloaded pair of deep-groove bearings.

Important: the envelope is not the same as the kinematic link length. A capsule link has a circular boss at each end and the joint axis sits at the boss centre, inset from the part end by the boss radius. Section 4 gives the true joint-to-joint lengths.

2. Assembly overview

Existing design — Robo_2_assembled.stl (envelope 155 × 260 × 213 mm, as-saved pose)



The as-printed assembly measures 155 × 260 × 213 mm in the pose it was saved in. Fully extended the chain is longer; section 4 gives the dimension chain that must total

450 mm.

3. Master parts list

Part	File	Group	Function	Envelope (mm)	Wall	Mass est.	Status
A-01	base_p1	Base	Lower base shell, D-profile	72.9 × 56.0 × 43.0	2.7	17.3 g	keep
A-02	base_p2	Base	Base cover / floor plate	72.9 × 56.0 × 18.0	2.8	8.8 g	keep
A-03	Motor_mount_base	Base	J1 servo mount, top-hat flange	46.0 × 46.0 × 33.0	2.3	4.9 g	revise
B-01	link1_base	Link	Upper-arm shell, lower half	147.5 × 47.5 × 29.0	2.0	15.8 g	LENGTHEN
B-02	link1_cover	Link	Upper-arm shell, upper half	196.5 × 46.5 × 33.2	2.2	22.1 g	MISMATCH
B-03	link2_base	Link	Forearm shell, lower half	147.5 × 47.5 × 29.0	2.0	15.8 g	LENGTHEN
B-04	link2_cover	Link	Forearm shell, upper half	146.5 × 46.5 × 33.2	1.9	14.8 g	LENGTHEN
B-05	link1_base_1.1	Link	Upper-arm shell rev 1.1 (longer)	256.0 × 56.0 × 34.5	2.7	43.2 g	REJECT
B-06	link1_cover 1.1	Link	Upper-arm cover rev 1.1	256.0 × 56.0 × 33.0	2.1	30.9 g	REJECT
B-07	link2_cover 2.1	Link	Forearm cover rev 2.1	147.5 × 47.5 × 35.8	1.9	17.5 g	keep
C-01	J2_p1	Joint	J2 shoulder yoke, driven side	80.0 × 64.8 × 85.5	—	—	revise

Part	File	Group	Function	Envelope (mm)	Wall	Mass est.	Status
C-02	J2_p2	Joint	J2 shoulder yoke, idler side	40.0 × 64.8 × 66.0	2.0	8.3 g	revise
C-03	J4_p1	Joint	J4 yoke, driven side	63.8 × 57.0 × 79.8	—	—	revise
C-04	J4_p2	Joint	J4 yoke, idler side	35.7 × 57.0 × 61.7	2.2	7.7 g	revise
C-05	Motor_housing_new	Joint	Elbow motor housing	55.5 × 53.0 × 69.0	2.6	17.4 g	keep
D-01	wrist_p1	Wrist	Wrist body, driven side	58.2 × 62.7 × 72.2	2.5	12.5 g	keep
D-02	wrist_p2	Wrist	Wrist body, idler side	31.7 × 62.7 × 58.2	2.2	7.0 g	keep
E-01	Motor_mount_J1	Mount	J1 servo mount	40.0 × 40.0 × 24.0	2.1	3.1 g	REDESIGN
E-02	Motor_mount_J2	Mount	J2 servo mount	46.0 × 46.0 × 38.2	2.3	5.4 g	REDESIGN
E-03	Motor_mount_wrist	Mount	Wrist servo mount	40.0 × 40.0 × 23.0	2.1	3.0 g	REDESIGN
E-04	Motor_mount	Mount	Generic servo mount	40.0 × 40.0 × 28.0	2.2	3.6 g	REDESIGN
E-05	Motor_mount_changed_ 2mm_length	Mount	Servo mount, +2 mm variant	40.0 × 40.0 × 30.0	2.4	4.1 g	REDESIGN
F-01	Motor_fixer_j2_p1	Fixer	J2 servo clamp, 6 mm plate	6.0 × 41.0 × 36.0	1.7	0.9 g	FAILS

Part	File	Group	Function	Envelope (mm)	Wall	Mass est.	Status
F-02	Motor_fixer_j2_p2	Fixer	J2 servo clamp, 4 mm plate	4.0 × 50.0 × 36.0	1.4	0.6 g	FAILS
F-03	motor fixer	Fixer	Servo clamp, U-channel	64.8 × 27.2 × 31.0	2.2	4.5 g	FAILS
F-04	motor fixer_wrist	Fixer	Wrist servo clamp	57.0 × 27.2 × 31.0	2.1	4.2 g	FAILS
G-01	ST3215	COTS	Feetech ST3215 serial-bus servo	45.2 × 37.8 × 24.7	4.3	23.4 g	COTS
G-02	thrust ball bearing	COTS	Thrust ball bearing Ø42	42.0 × 3.7 × 42.0	—	—	REPLACE
G-03	pin	Tool	Grasp pin / end effector	47.5 × 90.0 × 23.0	2.9	9.9 g	keep

Mass estimates assume PLA at roughly 35 % infill (0.55 g/cm³ effective) and are indicative only — weigh the real parts to confirm. Wall is the effective shell thickness computed as 2V/A; it is meaningless for solid parts and is shown as a dash where the mesh is not watertight.

4. Kinematic dimension chain — the 450 mm budget

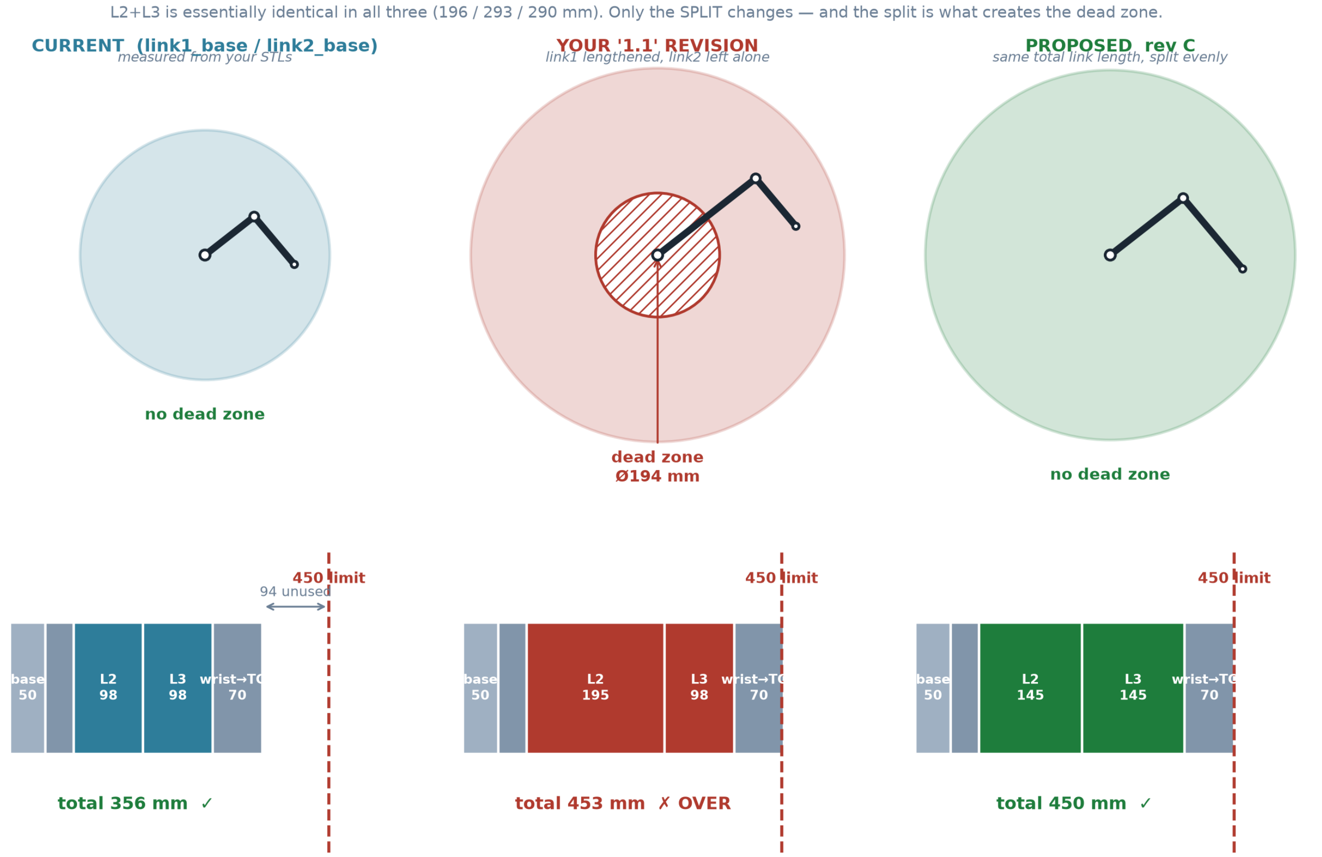
Segment	Length	Notes
Round base height	50	user's sketch dimension
Base top → J2 shoulder axis	40	shoulder barrel centre
L2 upper arm, J2 → J3	145	must equal L3
L3 forearm, J3 → J5	145	must equal L2
Wrist centre → TCP	70	J6 flange plus tool
TOTAL	450	meets the requirement exactly

Horizontal reach from the J1 axis is 145 + 145 + 70 = 360 mm.

Why L2 must equal L3

The reachable region of a two-link chain is an annulus whose inner radius is |L2 − L3|. Equal links collapse that inner hole to zero. Unequal links punch a dead zone directly in front of the robot that no amount of joint travel can reach.

Your two link generations vs the fix — measured from ~/Desktop/Robotic_arm_design/*.stl



Configuration	L2	L3	Dead zone	Total length
As printed	97.9	97.9	none	356 mm
Revision 1.1	195.1	97.9	Ø194 mm	453 mm — over
Recommended	145	145	none	450 mm

5. Measured link geometry

Joint centres were recovered by fitting circles to the end bosses (measure_joint_centres.py), because the bounding box overstates the kinematic length by the sum of the two boss radii.

Part	Envelope length	Boss radius A	Boss radius B	Joint-to-joint
link1_base	147.5	25.2	25.0	97.9
link2_base	147.5	25.2	25.0	97.9
link1_base_1.1	256.0	29.5	29.2	195.1
link1_cover	196.5	16.8	17.6	167.6
link2_cover	146.5	18.2	17.2	115.9
link1_cover1.1	256.0	27.1	27.1	196.0

Flag: link1_cover (envelope 196.5) does not pair with link1_base (envelope 147.5).

link2_cover (146.5) does pair with link2_base (147.5). Check this before reprinting — a mismatched shell pair is a likely cause of both the assembly gaps and any URDF export errors.

6. Link cross-section

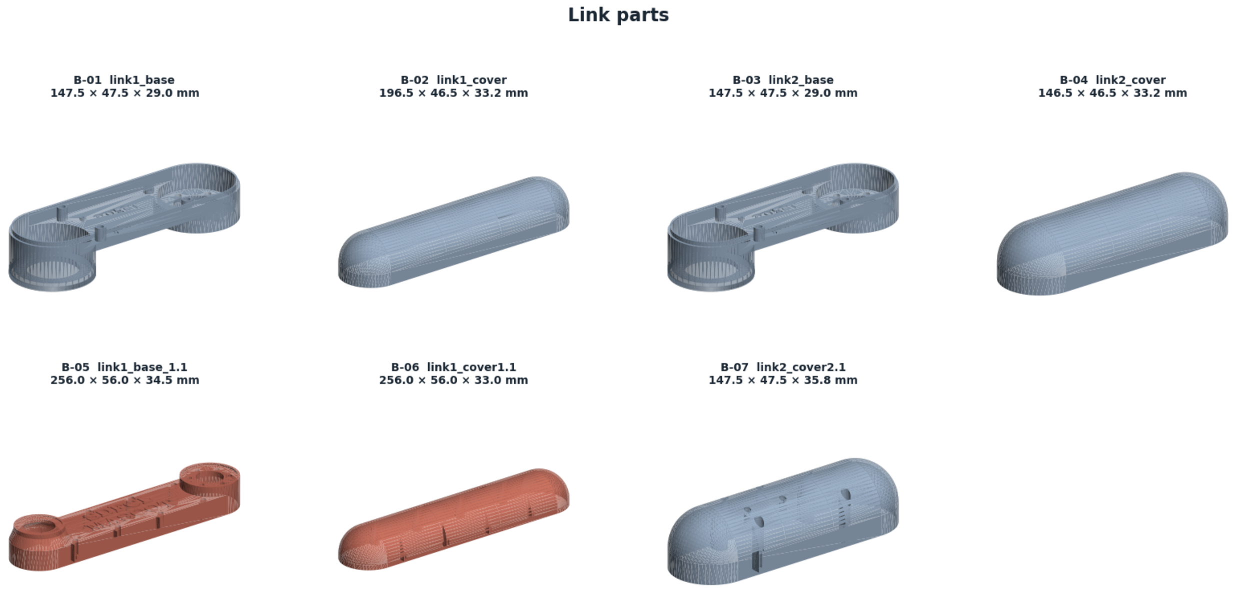
Property	Value
Outer width × depth	47.5 × 29.0
Wall thickness, as printed	2.0
Wall thickness, recommended	2.4 (6 × 0.4 mm nozzle passes)
Second moment of area I, closed, t=2.0	39,899 mm^4
Torsion constant J, closed, t=2.0	83,267 mm^4
Torsion constant J, seam open	661 mm^4 — a 126× loss

7. Part drawings by group

Base

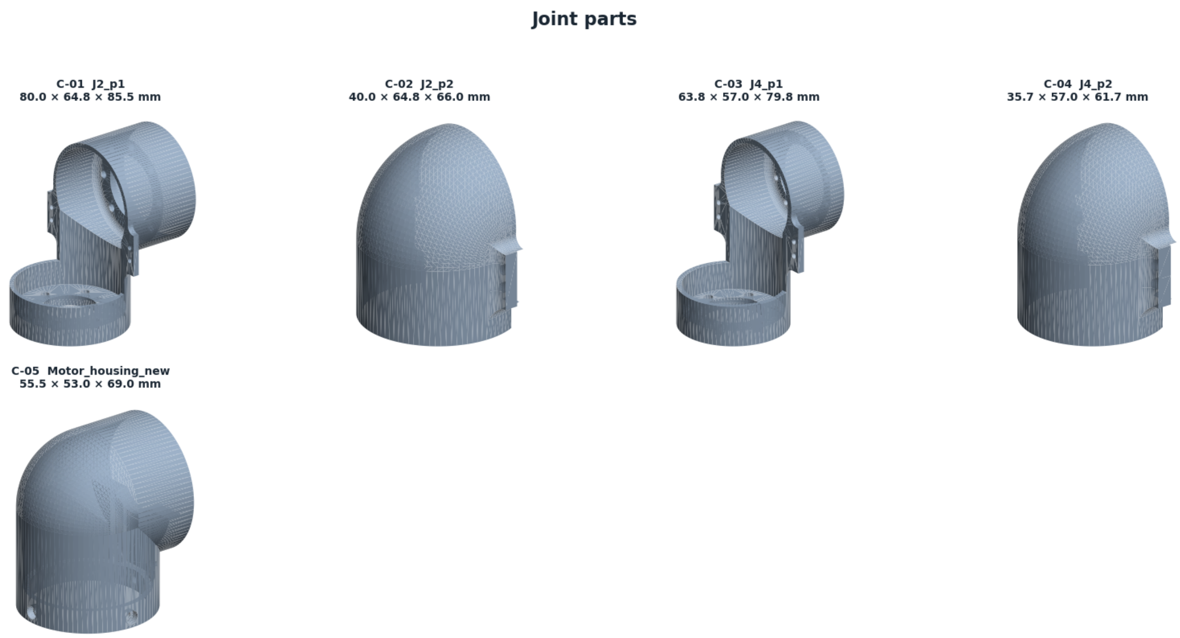
Part	Envelope (mm)	Wall	Status
A-01 base_p1	72.9 × 56.0 × 43.0	2.7	keep
A-02 base_p2	72.9 × 56.0 × 18.0	2.8	keep
A-03 Motor_mount_base	46.0 × 46.0 × 33.0	2.3	revise

Link



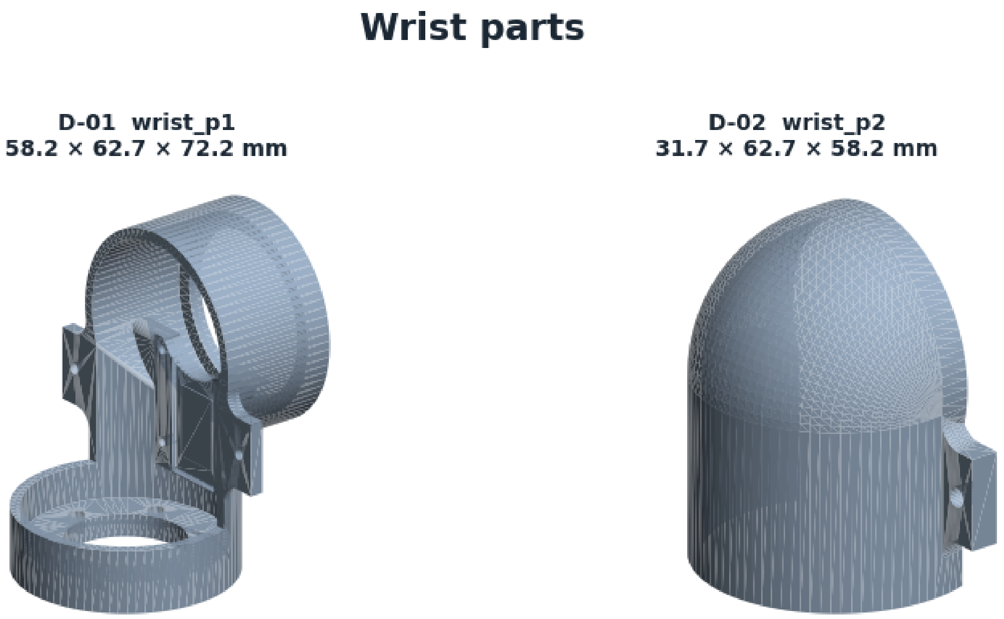
Part	Envelope (mm)	Wall	Status
B-01 link1_base	147.5 × 47.5 × 29.0	2.0	LENGTHEN
B-02 link1_cover	196.5 × 46.5 × 33.2	2.2	MISMATCH
B-03 link2_base	147.5 × 47.5 × 29.0	2.0	LENGTHEN
B-04 link2_cover	146.5 × 46.5 × 33.2	1.9	LENGTHEN
B-05 link1_base_1.1	256.0 × 56.0 × 34.5	2.7	REJECT
B-06 link1_cover1.1	256.0 × 56.0 × 33.0	2.1	REJECT
B-07 link2_cover2.1	147.5 × 47.5 × 35.8	1.9	keep

Joint



Part	Envelope (mm)	Wall	Status
C-01 J2_p1	80.0 × 64.8 × 85.5	—	revise
C-02 J2_p2	40.0 × 64.8 × 66.0	2.0	revise
C-03 J4_p1	63.8 × 57.0 × 79.8	—	revise
C-04 J4_p2	35.7 × 57.0 × 61.7	2.2	revise
C-05 Motor_housing_new	55.5 × 53.0 × 69.0	2.6	keep

Wrist

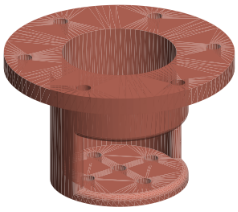


Part	Envelope (mm)	Wall	Status
D-01 wrist_p1	58.2 × 62.7 × 72.2	2.5	keep
D-02 wrist_p2	31.7 × 62.7 × 58.2	2.2	keep

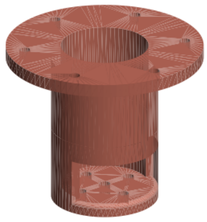
Mount

Mount parts

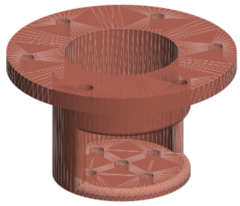
E-01 Motor_mount_J1
40.0 × 40.0 × 24.0 mm



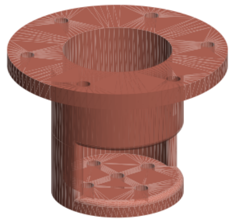
E-02 Motor_mount_J2
46.0 × 46.0 × 38.2 mm



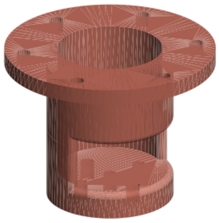
E-03 Motor_mount_wrist
40.0 × 40.0 × 23.0 mm



E-04 Motor_mount
40.0 × 40.0 × 28.0 mm



E-05 Motor_mount_changed_ 2mm_length
40.0 × 40.0 × 30.0 mm

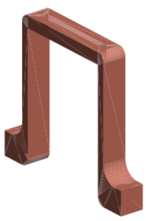


Part	Envelope (mm)	Wall	Status
E-01 Motor_mount_J1	40.0 × 40.0 × 24.0	2.1	REDESIGN
E-02 Motor_mount_J2	46.0 × 46.0 × 38.2	2.3	REDESIGN
E-03 Motor_mount_wrist	40.0 × 40.0 × 23.0	2.1	REDESIGN
E-04 Motor_mount	40.0 × 40.0 × 28.0	2.2	REDESIGN
E-05 Motor_mount_changed_ 2mm_length	40.0 × 40.0 × 30.0	2.4	REDESIGN

Fixer

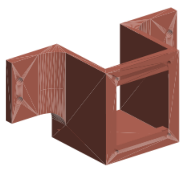
Fixer parts

F-01 Motor_fixer_j2_p1
6.0 × 41.0 × 36.0 mm

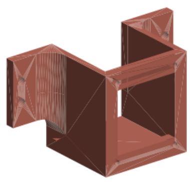


F-02 Motor_fixer_j2_p2
4.0 × 50.0 × 36.0 mm

F-03 motor fixer
64.8 × 27.2 × 31.0 mm



F-04 motor fixer_wrist
57.0 × 27.2 × 31.0 mm

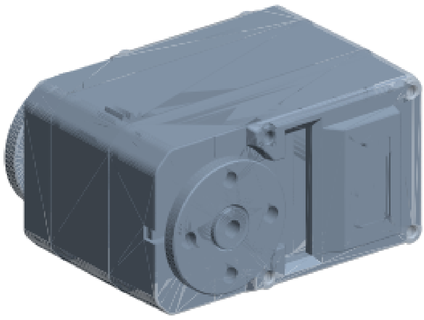


Part	Envelope (mm)	Wall	Status
F-01 Motor_fixer_j2_p1	6.0 × 41.0 × 36.0	1.7	FAILS
F-02 Motor_fixer_j2_p2	4.0 × 50.0 × 36.0	1.4	FAILS
F-03 motor fixer	64.8 × 27.2 × 31.0	2.2	FAILS
F-04 motor fixer_wrist	57.0 × 27.2 × 31.0	2.1	FAILS

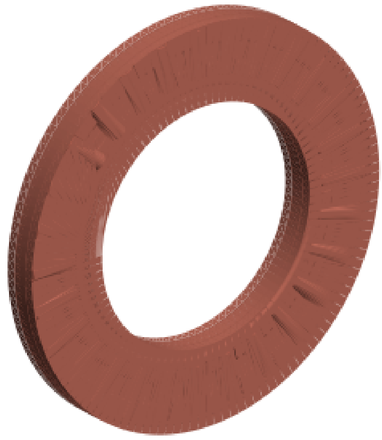
COTS

COTS parts

G-01 ST3215
45.2 × 37.8 × 24.7 mm



G-02 thrust ball bearing
42.0 × 3.7 × 42.0 mm

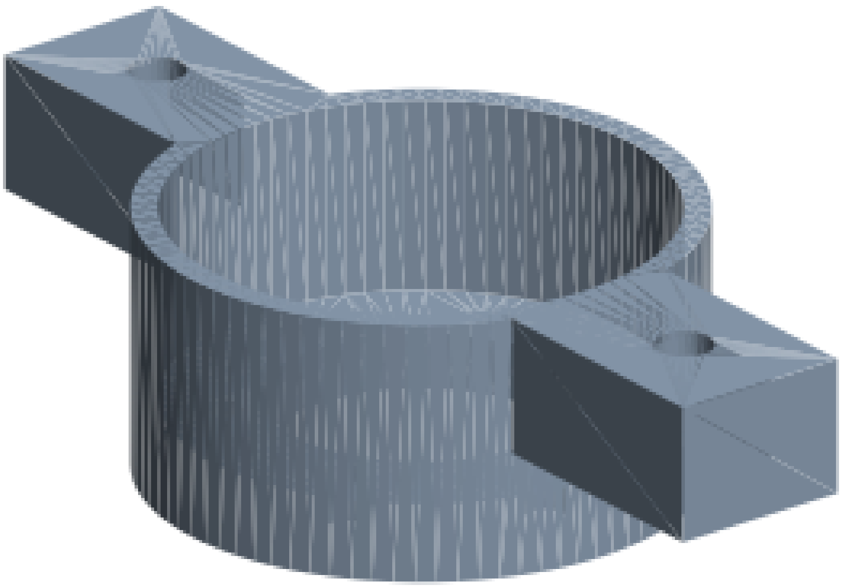


Part	Envelope (mm)	Wall	Status
G-01 ST3215	45.2 × 37.8 × 24.7	4.3	COTS
G-02 thrust ball bearing	42.0 × 3.7 × 42.0	—	REPLACE

Tool

Tool parts

G-03 pin
47.5 × 90.0 × 23.0 mm



Part	Envelope (mm)	Wall	Status
G-03 pin	47.5 × 90.0 × 23.0	2.9	keep

8. Critical interface dimensions

These are the dimensions that decide whether the arm assembles tightly. They matter more than any outside dimension.

Interface	Current	Specify	Reason
Bearing pocket, Ø42 bearing	Ø42.0 modelled	Ø42.15 press fit	FDM pockets print 0.1–0.4 mm undersize
Bearing pocket chamfer	none	0.5 × 45°	bearing starts square
Bearing axial seat	flat face	turned shoulder	flat printed faces are not flat

Interface	Current	Specify	Reason
Bearing spacing per joint	~5 (stacked)	≥ 40	moment stiffness ∝ spacing^2
M3 heat-set insert hole	—	Ø4.0	for a Ø4.6 × 5.8 insert
Insert boss outer diameter	—	≥ Ø9.0	≥ 2.2 mm wall or it splits
Seam fastener pitch	—	≤ 25	keeps the section closed
Seam interlock lip	none	1.5 mm tongue-and-groove	shear in bearing, not friction
Internal fillet radius	~0 sharp	≥ 2.0	drops Kt from ≥3 to ~1.3
Servo clamp form	4–6 mm bracket	full-perimeter collar, 4 mm wall	carries torque in shear, not bending

Insert dimensions are typical values — check your specific insert's datasheet.

9. Bill of materials — non-printed

Item	Qty	Note
Feetech ST3215 servo (30 kgf·cm)	5	J1, J3, J4, J5, J6
Feetech ST3250 servo (50 kgf·cm)	1	J2 shoulder — needs the extra torque
Deep-groove ball bearings	12	2 per joint, spaced ≥ 40 mm
M3 heat-set inserts, Ø4.6 × 5.8	~40	seam and mounts
M3 socket-head screws, 8–16 mm	~40	
M3 preload screw + spacer per joint	6	removes bearing clearance

Bearing sizes are not yet fixed — they depend on the shaft diameter chosen for each joint. See the open questions in the analysis report.